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PRE QUARTERLY EXAMINATION - 2025

CLASS : 12
Time : 3.00 Hours

CHEMISTRY

REG. No
Maximum Marks : 70

PART - I

I **Choose the correct answer.** 15x1=15

1. Zinc is obtained from Zno by
   a) Carbon Reduction b) Reduction using silver
   c) Electrochemical process d) Acid leaching
2. Which of the metal is extracted by Hall-Heroult process?
   a) Al b) Ni c) Cu d) Zn
3. Which of the following is not Sp2 hybridised?
   a) Graphite b) Graphene c) Fullerene d) Dry ice
4. The element that does not show catenation among the following P-block element is
   a) Carbon b) Silicon c) Lead d) Germanium
5. Which of the following is strongest acid among all?
   a) HI b) HF c) HBr d) Hcl
6. The most common oxidation state of actinoids is
   a) +2 b) +3 c) +4 d) +6
7. The Vacant space in bcc lattice unit cell is
   a) 48% b) 23% c) 32% d) 26%
8. The Crystal with a metal deficiency defect is
   a) Nacl b) Feo c) Zno d) Kcl
9. The addition of a Catalyst during a chemical reaction alters which of the following quantities?
   a) Enthalpy b) Activation energy c) Entropy d) Internal energy
10. The rate constant of a reaction is $5.8 \times 10^{-2} S^{-1}$. The order of the reaction is
    a) First order b) Zero order c) Second order d) Third order
11. Which of these is not likely to act as lewis base?
    a) BF_3 b) PF_3 c) CO d) F^-
12. The p^H of 10^{-5} M KOH solution will be
    a) 9 b) 5 c) 19 d) none of these
13. Which of the following compound can be used as antifreeze in automobile radiators?
    a) Methanol b) Ethanol c) Neopentyl alcohol d) Ethan-1,2-diol
14. $HO\text{-}CH_2\text{-}CH_2\text{-}OH$ on heating with periodic acid gives
    a) Methanoic acid b) Glyoxal c) Methanal d) CO_2
15. The formation of cyanohydrin from acetone is an example of
    a) Nucleophilic Substitution b) Electrophilic Substitution
    c) Electrophilic addition d) Nucleophilic addition

PART - II

II **Answer any 6 questions. Question No : 24 is compulsory.** 6x2=12

16. What are the various steps involved in extraction of pure metals from their ores?
17. How will you convert boric acid to boron nitride?

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18. What is inert pair effect?
19. Which is more Stable? Fe^{3+} (or) Fe^{2+}. Explain.
20. Calculate the number of atoms in a fcc unit cell.
21. What are Lewis acids and bases? Give two examples for each.
22. Explain Kolbe's reaction.
23. Write a note on Benzoin Condensation>?
24. Show that in case of first order reaction the time required for 99.9% completion is nearly ten times the time required fro half completion of the reaction.

PART - III

III Answer any 6 questions. Question No: 33 is compulsory. **6x3=18**

25. What is the difference between minerals and ores?
26. Write a note on Fisher tropsch synthesis.
27. Give the uses of Helium.
28. What are Interstitial Compounds?
29. Give any three characteristics of Ionic Crystals.
30. Write Arrhenius equation and explain the terms involved.
31. What is common ion effect? Give example.
32. Write the preparation of ethylene glycol.
33. Write the IUPAC name of the following.
    (a) CH_2=CH-CHO (b) CH_3-CH=CH-CHO (c) $C_6H_5COCH_3$

PART - IV

IV Answer the following questions. **5x5=25**

34. Explain zone refining process with an example. [OR]
    (a) Give the uses of Borax. (3)
    (b) Write a short note on Hydroboration. (2)
35. (a) What are Interhalogen compounds? Give example. (2)
    (b) Give a reason to support that sulphuric acid is a dehydrating agent. **[OR]** (3)
    What is lanthanide contraction? List the Consequences of lanthanide contraction.
36. (a) What is meant by the term "Coordination number"? What is the coordination number of atoms in a bcc structure? (2)
    (b) Write a note on Frenkel defect. **[OR]** (3)
    Derive integrated rate law for a zero order reaction A → Product.
37. (a) Define : pH (2)
    (b) Define : Ionic Product of water. Give its value at room temperature. **[OR]** (3)
    Derive an expression for Ostwald's dilution law.
38. Distinguish 1^0, 2^0, 3^0 alcohols using Lucas Test.

[OR]

Write the following reactions.
(i) Clemmensen Reduction (ii) Haloform reaction (iii) Knoevenagal reaction

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